An Observational Study of Transcoder Feature Attribution on Indirect Object Identification in GPT-2 Small*

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Abstract

We report an observational study of how transcoder-based feature attribution behaves when used to extract and causally test feature circuits in GPT-2 small (124M parameters, CPU-only). Using pretrained transcoders from `pchlenski/gpt2-transcoders` and Input×Gradient (IxG) attribution, we systematically observe the method along five dimensions: single-feature causal direction, cross-prompt stability of attributed features, transfer of frozen feature sets to paraphrases, cross-task specificity, and quantitative calibration of attribution scores against realized ablation effects. Our central observation is that simultaneously replacing all 12 MLP layers with their transcoder approximations produces a compounding 5.6-logit displacement of the task signal ($D: +4.15 \rightarrow -1.46$), which inverts the sufficiency metric and dominates feature-level effects. Within this regime we nonetheless observe consistent, interpretable structure: a single feature (L11, F15690) is the most attributed feature on 8 of 10 source prompts, the attributed feature set has negligible causal effect on an unrelated greater-than task, and the sign of attribution reliably predicts the direction of ablation effects even when its magnitude does not. We characterize the approximation-error mechanism quantitatively and discuss what it implies for the design of causal interpretability experiments that depend on full-MLP replacement.

1 Introduction

Mechanistic interpretability seeks to decompose the internal computations of neural networks into human-understandable components [Olah et al., 2020, Elhage et al., 2021]. One recurring object of study is the *feature circuit*: a sparse subset of internal features hypothesized to causally explain a model's behavior on a task, together with the claim that such a circuit generalizes across surface variations of that task. Whether this claim holds—and whether current tooling can measure it—is an empirical question.

The *Indirect Object Identification* (IOI) task is a standard probe for this question in transformer language models [Wang et al., 2022]. Given a sentence such as *"When Mary and John went to the store, John gave the bag to [BLANK]"*, a competent model completes with the indirect object (Mary) rather than the repeated subject (John). The logit difference $D = \text{logit(IO)} - \text{logit(S)}$ quantifies the strength of this preference.

Transcoders [Dunefsky et al., 2024] replace each MLP sublayer of a transformer with a sparse autoencoder trained to reconstruct the MLP's output from its input, decomposing MLP computation

*Code and data: `https://github.com/dzhan111/ioi_tc_pilot`

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into interpretable features. Each feature carries an Input×Gradient (IxG) attribution score c_n estimating its contribution to a target output direction. Together, transcoders and IxG attribution support feature-level causal hypotheses that can be tested by mean-ablating selected features and observing the change in D.

This paper is an *observational study*: rather than proposing a new technique, we systematically observe how the transcoder + IxG pipeline behaves on a task whose ground-truth circuit is independently well-characterized in prior work. We organize our observations around five research questions (Section 2.6), report the measured behavior of the method under each, and analyze the dominant mechanism—transcoder approximation error—that shapes the results. Our aim is to document, in reproducible detail, both where the method exhibits stable and interpretable structure and where its measured quantities cease to be meaningful, so that subsequent causal-interpretability designs can account for these effects.

2 Study Design

2.1 Model and Stimuli

We study GPT-2 small (12 layers, 768-dimensional residual stream, 12 attention heads, 124M parameters), run on CPU for hardware-independent reproducibility. All attribution and intervention passes use a fixed random seed (42); random baselines use independently fixed dataset-level seeds.

Our stimulus set comprises 10 IOI *source* prompts with varied name pairs and contexts (store, park, library, etc.), 5 IOI *paraphrase* prompts with fresh name pairs and the same grammatical structure, and 5 *greater-than* (GT) prompts (e.g. *"The war lasted from the year 18XX to the year 18[BLANK]"*) serving as a structurally unrelated control task.

2.2 Transcoders

We use the 12 pretrained GPT-2 small transcoders from `pchlenski/gpt2-transcoders`. Each is a sparse autoencoder with $d_{\text{sae}} = 24{,}576$ features hooking into the post-LayerNorm2 activation of its MLP layer (`blocks.{l}.ln2.hook_normalized`). The forward pass returns $(\hat{x}, f, \ldots)$ where $\hat{x} \in \mathbb{R}^{d_{\text{model}}}$ reconstructs the MLP output and $f \in \mathbb{R}^{d_{\text{sae}}}$ is the sparse feature-activation vector. All transcoders are loaded in CPU eval mode with device forced to CPU regardless of checkpoint metadata.

2.3 Attribution

For a prompt, layer ℓ, and output direction $\hat{r}$, the IxG score for feature n is

$$c_n = \left(W_{\text{dec}}^{(\ell)}[n] \cdot \hat{r}\right) \cdot f_n, \tag{1}$$

where $W_{\text{dec}}^{(\ell)}[n] \in \mathbb{R}^{d_{\text{model}}}$ is the n-th decoder column and f_n is feature n's activation at the final token. We use the true gradient $\hat{r} = \partial D / \partial \text{resid}_{11}$ via autograd, which accounts for the final-LayerNorm Jacobian, rather than the linear approximation $W_U[:, \text{IO}] - W_U[:, \text{S}]$.

2.4 Intervention Framework

To keep attribution and intervention in a single, consistent activation space, all 12 MLP layers are replaced by their transcoders during both the cache-building pass (for Eq. 1) and all ablation runs. Non-ablated layers use a passthrough wrapper returning $\hat{x}$ unmodified; ablated layers replace selected features with their dataset mean before decoding:

$$\hat{x}_{\text{ablated}} = \hat{x} + \sum_{n \in S} (\mu_n - f_n)\, W_{\text{dec}}[n], \tag{2}$$

where μ_n is the mean activation of feature n over a 50-sentence neutral corpus sampled at the final token position. We measure effects relative to the transcoder baseline $D_{\text{full}}^{(\text{TC})}$ rather than the original model's D, so that observed ablation effects are not confounded with the static reconstruction offset of the transcoders.

2.5 Measured Quantities

Logit difference D. $D = \mathrm{logit}(\mathrm{IO}) - \mathrm{logit}(\mathrm{S})$ at the final token.

Sufficiency, Faith$_S$. For a feature set S,

$$\mathrm{Faith}_S = \frac{D_{\text{full}} - D_S}{D_{\text{full}} - D_{\text{corrupt}}}, \tag{3}$$

with D_S the value after ablating S and $D_{\text{corrupt}} = 0$ (a neutral, no-preference reference). $\mathrm{Faith}_S = 1$ indicates S fully accounts for the behavior; $\mathrm{Faith}_S = 0$ indicates S is causally irrelevant.

Jaccard similarity. $J(A, B) = |A \cap B|/|A \cup B|$, used to measure overlap between feature sets extracted from different prompts.

Spearman ρ. Rank correlation between predicted c_n and realized per-feature ΔD, used to assess whether attribution magnitude tracks causal effect magnitude.

2.6 Research Questions

We organize our observations around five questions:

RQ1 (Direction). Does ablating an attributed feature move D in the direction its c_n predicts?

RQ2 (Stability). Are attributed feature sets consistent across prompts that share the IOI structure?

RQ3 (Transfer). Does a frozen feature set extracted from source prompts remain causally sufficient on unseen paraphrases?

RQ4 (Specificity). Is an IOI feature set causally inert on a structurally unrelated task?

RQ5 (Calibration). Does the magnitude of c_n predict the magnitude of the realized ablation effect?

3 Observations

3.1 RQ1 — Single-Feature Causal Direction

We extract the top feature by c_n on the first source prompt and ablate it in isolation, comparing $D^{(\text{TC})}_{\text{full}}$ against D_{ablated}. The top feature is (L11, F15690) with $c_n = 0.112$; a positive c_n predicts that removal lowers D.

Observation. Ablation yields $\Delta D = D^{(\text{TC})}_{\text{full}} - D_{\text{ablated}} = +0.0086$, matching the predicted sign. The realized effect is, however, only 7.7% of the predicted c_n, an early indication of substantial attenuation within the all-MLP transcoder regime (Section 4).

3.2 RQ2 — Cross-Prompt Stability of Attributed Features

We build IxG circuits on all 10 source prompts and measure pairwise Jaccard overlap between their top-20 feature sets, the fraction of prompt pairs sharing the same top-1 feature, and the consensus set S of features appearing in ≥ 2 circuits.

Table 1: Cross-prompt stability of attributed feature sets (10 IOI source prompts).

Quantity	Value
Top-20 Jaccard (median)	0.250
Top-20 Jaccard (mean)	0.296
Top-20 Jaccard (min / max)	0.111 / 0.739
Top-1 Jaccard (median)	**1.000**
Consensus set size $\|S\|$	45 features

Observation. The structure is bimodal (Table 1): a single feature, (L11, F15690), is the top-attributed feature on 8 of 10 prompts (top-1 Jaccard $= 1.0$ across the majority), while the surrounding top-20 set overlaps only $\sim$25% between prompts. The method thus isolates one highly stable feature embedded in a diffuse, prompt-specific tail. The 45-feature consensus set used in subsequent observations inherits this tail and is correspondingly noisy.

3.3 RQ3 — Transfer of a Frozen Feature Set to Paraphrases

We ablate the consensus set S (45 features) and a size-matched random baseline on 5 paraphrase prompts, measuring Faith_S relative to $D_{\text{corrupt}} = 0$.

Table 2: Sufficiency of the frozen feature set on paraphrase prompts.

Prompt	$D^{(\text{TC})}_{\text{full}}$	Faith_S(attr)	Faith_S(rand)
Grace / Mike (garden)	-1.144	-0.065	-0.000
Lily / Dan (library)	-1.922	$+0.002$	-0.000
Eva / Leo (hall)	-0.464	-0.127	-0.000
Nina / Carl (yard)	-1.317	-0.001	-0.000
Vera / Jack (field)	-2.687	$+0.018$	-0.000
Mean	-1.507	$\mathbf{-0.035}$	$\mathbf{0.000}$

Observation. Across all five paraphrases the transcoder baseline $D^{(\text{TC})}_{\text{full}}$ is negative (Table 2): under full-MLP replacement the model no longer prefers the indirect object. Because $D_{\text{full}} < D_{\text{corrupt}} = 0$, the denominator of Eq. 3 is negative and Faith_S no longer carries its intended meaning. The attributed set and the random baseline are statistically indistinguishable in this regime (gap $= -0.035$). We therefore read this not as evidence about transfer per se, but as evidence that full-MLP replacement does not preserve the task signal the metric presupposes (Section 4).

3.4 RQ4 — Cross-Task Specificity

We apply the IOI consensus set S to 5 greater-than prompts and measure Faith_S on the digit-ordering objective, where IOI features should be inert.

Table 3: Cross-task effect of the IOI feature set on greater-than prompts.

GT Prompt (years)	$D^{(\text{TC})}_{\text{full}}$	Faith_S
$1815 \rightarrow 18$	-0.280	-0.003
$1823 \rightarrow 18$	-0.256	$+0.081$
$1847 \rightarrow 18$	$+0.254$	$+0.006$
$1862 \rightarrow 18$	-0.147	-0.024
$1878 \rightarrow 18$	$+0.173$	$+0.180$
Mean	-0.051	$\mathbf{+0.048}$

Observation. The mean cross-task Faith_S is 0.048, near zero (Table 3). Ablating the IOI-derived set leaves the greater-than objective essentially undisturbed. This indicates that the features the method selects are tied to IOI structure rather than to generic high-activation directions that would perturb any task—a property that remains observable despite the metric's degradation on the IOI task itself.

3.5 RQ4a — Attribution versus Activation Magnitude

As a companion observation to RQ4, we compare the attributed set S against a size-matched set chosen purely by activation magnitude (ignoring gradient alignment) on the paraphrase prompts.

Observation. Mean Faith_S is -0.035 for attribution and -0.032 for magnitude (gap $= -0.002$); the two are indistinguishable. As in RQ3, this comparison is made on prompts where $D^{(\text{TC})}_{\text{full}} < 0$, so it characterizes the degraded regime rather than the relative merit of the two selection rules; it does not, on its own, support a conclusion about whether gradient alignment adds information.

3.6 RQ5 — Quantitative Calibration

We ablate each of the 45 features in S individually across all 10 source prompts ($n = 450$ pairs) and correlate the predicted c_n with the realized ΔD.

Observation. Spearman $\rho = -0.018$ ($p = 0.70$): there is no monotonic relationship between attribution magnitude and realized ablation magnitude, even on the source prompts from which the attribution was derived. The directionally correct but heavily attenuated single-feature effect from RQ1 does not extend to a usable rank ordering across features. This is consistent with attenuation that is feature- and path-dependent rather than uniform (Section 4); uniform attenuation would preserve ρ.

4 Mechanism: Compounding Transcoder Approximation Error

The observations under RQ3 and RQ5 share a common origin, which we characterize directly here.

Table 4: Effect of transcoder replacement on the IOI logit difference.

Model Configuration	D	ΔD from original
Original GPT-2 (all MLPs)	$+4.15$	—
All 12 MLPs replaced by TCs	-1.46	-5.61
TC baseline on paraphrase prompts		
Grace / Mike	-1.14	—
Lily / Dan	-1.92	—
Eva / Leo	-0.46	—
Nina / Carl	-1.32	—
Vera / Jack	-2.69	—

Each transcoder is trained independently to approximate its layer's MLP output given activations drawn from the *original* model. When all 12 are chained, the residual stream at each layer drifts from the distribution the next transcoder was trained on. This distributional shift compounds across depth: a small error at layer 0 moves the layer-1 input off-manifold, amplifying the error at layer 1, and so on. The cumulative result is a 5.6-logit displacement (Table 4)—larger in magnitude than the original 4.15-logit IOI signal itself.

The consequence for the sufficiency metric is direct. With $D_{\text{full}}^{(\text{TC})} < 0 < D_{\text{corrupt}}$, Eq. 3 acquires a negative denominator,

$$\text{Faith}_S = \frac{D_{\text{full}} - D_S}{D_{\text{full}} - 0}, \quad D_{\text{full}} < 0, \tag{4}$$

so any ablation that nudges D toward zero registers as negative Faith_S— inverting the intended reading. At the single-feature level the same mechanism manifests as attenuation: the realized effect in RQ1 (0.0086) is 7.7% of the predicted c_n (0.112). Because the off-manifold error is feature- and path-specific rather than a uniform scaling, the attenuation also scrambles the cross-feature rank ordering observed in RQ5.

5 What Remains Stable

Three properties persist despite the metric degradation and are, we argue, the substantive findings of this study.

A single dominant, interpretable feature. Feature (L11, F15690) is the top-attributed feature on 8 of 10 source prompts, with consistently positive c_n in the range 0.039–0.136 across diverse name pairs and contexts. Its prominence and consistency suggest it encodes a grammatical property of the IOI construction (plausibly the identification of the non-repeated name as the indirect object), and it survives as a stable signal even where aggregate metrics fail.

Task-specific selection. The attributed feature set has near-zero causal effect on the greater-than task ($\text{Faith}_S = 0.048$). The method is not merely surfacing globally high-activation directions; its selections are tied to the IOI task structure.

Reliable attribution direction. The sign of c_n correctly predicts the direction of the single-feature ablation effect. Direction is recoverable from the attribution even where magnitude is not.

6 Implications for Causal Interpretability Designs

The observations above bear on any causal-interpretability protocol that relies on full-MLP transcoder replacement.

Signal preservation is a precondition, not an assumption. A sufficiency metric such as Faith_S is only interpretable when the intervention framework preserves the behavior it measures. Our results show this cannot be taken for granted: for IOI in GPT-2 small, full-MLP replacement violates it outright. Designs should report the framework's task signal ($D^{(\mathrm{TC})}_{\mathrm{full}}$ vs. original D) as a precondition check before interpreting downstream causal metrics.

Localized replacement. Restricting transcoder replacement to the single most relevant layer (here, layer 11, which contains the dominant feature) would bound the approximation error to roughly one layer rather than compounding it across twelve, and is the most direct route to a regime in which $D^{(\mathrm{TC})}_{\mathrm{full}} > 0$.

Hook-based intervention on residual-stream SAEs. Sparse autoencoders trained on residual-stream positions, used with hook-based clamping rather than computation-replacing wrappers, decouple feature observation from model behavior entirely and avoid the compounding-error mechanism we document.

Narrower feature sets. The bimodal stability we observe (top-1 Jaccard 1.0, top-20 Jaccard 0.25) suggests that consensus sets built from a small k (5–10) would be purer and more likely to behave consistently than the diffuse top-20 sets used here.

7 Summary of Observations

Table 5: Summary of observations across the five research questions.

RQ	Dimension	Key Quantity	Value	Observation
1	Direction	sign(ΔD) vs. sign(c_n)	match	directionally correct
2	Stability	top-1 / top-20 Jaccard	1.00 / 0.25	one stable feature
3	Transfer	$D^{(\mathrm{TC})}_{\mathrm{full}}$ (mean)	-1.51	signal not preserved
4	Specificity	cross-task Faith_S	$+0.05$	task-specific
5	Calibration	Spearman ρ	-0.02	no magnitude tracking

8 Conclusion

We conducted an observational study of transcoder + IxG feature attribution on the IOI task in GPT-2 small, characterizing the method along five dimensions. The dominant phenomenon is a compounding 5.6-logit approximation error under full-MLP replacement that displaces the task signal and renders the sufficiency metric uninterpretable for transfer and calibration. Against this backdrop the method still exhibits stable, interpretable structure: a single dominant IOI feature consistent across prompts, task-specific feature selection, and reliable attribution direction at the single-feature level.

The broader observation is methodological. Transcoder frameworks that replace whole computational sublayers should be characterized for task-signal preservation before their causal metrics are interpreted, and localized or hook-based interventions are the natural way to obtain a regime in which those metrics are meaningful. We document the present regime in full so that such designs can be built on measured rather than assumed behavior.

Reproducibility

All code, results, and the LaTeX source for this paper are available at `https://github.com/dzhan111/ioi_tc_pilot`. The full pipeline runs on CPU in approximately 45 minutes; transcoder checkpoints download automatically from `pchlenski/gpt2-transcoders` on HuggingFace. Environment: Python 3.9, PyTorch 2.2.0 (CPU), TransformerLens 1.11.0.

- `scripts/00_smoke.py`–`05_calibration.py`: reproduce the observations for RQ1–RQ5
- `results/*.json`: raw numeric outputs
- `src/pilot/metrics.py`: Faith$_S$, Jaccard, Spearman ρ
- `tests/`: 17 unit tests covering the metric implementations

References

Dunefsky, J., Chlenski, P., and Nanda, N. Transcoders find interpretable LLM feature circuits. *arXiv preprint arXiv:2406.11944*, 2024.

Elhage, N., Nanda, N., Olsson, C., Henighan, T., Joseph, N., Mann, B., Askell, A., Bai, Y., Chen, A., Conerly, T., et al. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 2021.

Olah, C., Cammarata, N., Schubert, L., Goh, G., Petrov, M., and Carter, S. Zoom in: An introduction to circuits. *Distill*, 5(3):e00024–001, 2020.

Wang, K., Variengien, A., Conmy, A., Shlegeris, B., and Steinhardt, J. Interpretability in the wild: A circuit for indirect object identification in GPT-2 small. *arXiv preprint arXiv:2211.00593*, 2022.